Quasinormal modes of Schwarzschild anti-de Sitter black holes: a spectral analysis

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We study quasinormal modes (QNMs) of a minimally coupled massless scalar field on four-dimensional Schwarzschild–anti-de Sitter black holes using a Chebyshev spectral method. After compactifying the exterior domain, the radial problem is formulated as a quadratic matrix pencil in the dimensionless frequency. For the standard Dirichlet, or vanishing-field, boundary condition at the conformal AdS boundary, we reproduce the known scalar spectra across small, intermediate, and large black holes, including long overtone sequences and the expected approach to pure-AdS normal modes in the small black hole limit. We then deform the AdS boundary condition by imposing a generalized relation between the two independent asymptotic coefficients of the massless scalar. This deformation is treated as a generalized coefficient boundary condition for the massless scalar, not as the usual alternative quantization for scalars in the Breitenlohner–Freedman window. The Dirichlet endpoint recovers the stable standard spectrum. For every non-Dirichlet value examined, and for representative small, intermediate, and large black holes, we find an additional mode with positive imaginary part, signaling a boundary condition induced instability. A near-Dirichlet refinement finds no finite critical angle down to the smallest deformation probed.

I. INTRODUCTION

QNMs are the characteristic damped oscillations of black holes. They encode how perturbations relax after a disturbance and provide a coordinate-invariant description of the dissipative part of black hole dynamics. In asymptotically flat spacetimes, the standard QNM problem is defined by imposing purely ingoing behavior at the event horizon and purely outgoing behavior at spatial infinity. In asymptotically de Sitter spacetimes, the role of spatial infinity is replaced by the cosmological horizon, where one again imposes an outgoing radiative condition. In both cases, the boundary conditions are radiative at both ends of the physical domain, and the resulting eigenvalue problem is intrinsically non-self-adjoint, leading to a discrete set of complex frequencies. Comprehensive reviews of the role of QNMs in black hole physics can be found in [1–3]. The situation is qualitatively different in asymptotically anti-de Sitter (AdS) spacetimes. The conformal boundary of AdS is timelike, and hence, the large radius endpoint is not a wave zone. There is no canonical outgoing wave condition at infinity. Instead, one must prescribe boundary data at the AdS boundary. Thus, in AdS, the choice of boundary condition is not a technical detail added after the differential equation has been written down. It is part of the definition of the spectral problem itself. This feature is central both mathematically and physically. On the mathematical side, AdS is not globally hyperbolic, and the evolution problem is well posed only after an admissible boundary condition has been specified at the conformal boundary. [4, 5] formulated this issue in terms of self-adjoint extensions of the spatial part of the wave operator where different extensions correspond to different allowed boundary conditions at infinity. On the physical side, in the context of gauge/gravity duality, AdS QNM frequencies determine relaxation timescales of perturbations in the dual field theory, so the boundary condition at infinity fixes which source-response problem is being studied [2, 6]. The AdS QNM problem is, however, conceptually distinct from the self-adjoint evolution problem. The boundary condition at the conformal boundary may be chosen from the class of admissible boundary conditions, but the event horizon condition is not a regularity condition for a self-adjoint spatial operator. It is a radiative condition because QNMs are required to be purely ingoing at the black hole horizon. This replacement is what makes the QNM boundary value problem non-Hermitian and gives rise to complex frequencies. In this sense, the AdS-boundary condition fixes the domain of the problem at infinity, while the ingoing horizon condition encodes dissipation through the black hole.

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The first systematic numerical study of scalar QNMs of Schwarzschild–AdS black holes was carried out by [6], who imposed purely ingoing behavior at the event horizon and a vanishing field condition at the AdS boundary. Their power series method became the standard benchmark for scalar perturbations in SAdS. Subsequent studies extended the analysis to other fields, different black hole sizes, and higher overtones. [7] studied the small black hole regime and showed that the scalar modes approach the normal modes of pure AdS as the horizon radius tends to zero. [8] investigated the scalar, electromagnetic and gravitational spectra in detail, including the asymptotic large overtone structure. Further numerical checks and alternative algorithms include the asymptotic iteration method of [9], the interpolation matrix method of [10], and the continued fraction calculation of [11].

Boundary conditions in AdS become still subtler for fields beyond the minimally coupled scalar. For Maxwell perturbations, a field-vanishing condition is reflective, but it is not the most general condition associated with vanishing energy flux at the AdS boundary. Imposing vanishing energy flux leads to two possible Robin branches, one reproducing the usual spectrum found in earlier calculations and another giving an additional branch of electromagnetic QNMs [12, 13]. For gravitational perturbations, the correct boundary condition depends on the sector and on the physical prescription imposed at the boundary. In particular, fixed boundary metric or AdS/CFT-motivated conditions lead naturally to parity dependent boundary conditions where the axial sector may be treated with a Dirichlet-type condition, while the polar sector is associated with a Robin-type condition when the boundary metric is kept fixed [14, 15]. These examples show that the QNM spectrum of an AdS black hole is not a single object until the boundary condition at infinity has been specified.

The present work focuses on the scalar sector of four-dimensional Schwarzschild–AdS black holes. Our first aim is to construct and validate a high-precision Chebyshev spectral implementation for the standard minimally coupled massless scalar with the Dirichlet, or vanishing-field, condition at the conformal AdS boundary. This problem provides a stringent test because a large body of independent numerical data is available. We formulate the radial equation as a quadratic matrix pencil in the dimensionless frequency $\Omega = \omega R$, compactify the exterior domain, factor out the asymptotic QNM behavior, and identify physical modes by matching spectra computed at three different spectral resolutions. We compare the resulting frequencies with the Horowitz–Hubeny power-series method, Konoplya's small black hole data, the Cardoso–Konoplya–Lemos overtone tables, the asymptotic iteration method, the Lin–Qian matrix method, and the continued fraction calculation of Daghigh, Green and Morey. Our second aim is to use the same spectral framework to probe how the scalar spectrum changes when the standard Dirichlet condition is deformed. For the minimally coupled massless scalar, the usual finite-energy prescription at the AdS boundary selects the Dirichlet branch. By contrast, a genuine Robin family in the standard alternative-quantization sense is naturally associated with scalar fields for which both asymptotic falloffs are admissible, such as fields in the Breitenlohner–Freedman window [5, 16]. Here, we pursue a different question, namely we keep the same minimally coupled massless scalar field, but impose a generalized coefficient relation between the two independent asymptotic data in its large-radius expansion. The parameter controlling this relation interpolates between the standard Dirichlet endpoint and a coefficient-Neumann endpoint, with intermediate values defining a controlled deformation of the scalar QNM problem. This deformation should not be confused with the usual Robin boundary condition arising in alternative quantization. Rather, it should be viewed as a generalized source-response relation for the minimally coupled massless scalar. It allows us to ask whether the stability of the standard scalar SAdS spectrum is robust under non-Dirichlet deformations of the boundary data. Related Robin-type AdS boundary conditions and their dynamical consequences have been studied in different scalar settings in [16, 17]. The main result of this second analysis is that the generalized coefficient condition generically produces an unstable mode. The Dirichlet endpoint is stable and reproduces the standard scalar spectrum, but for every nonzero value of the deformation parameter θ tested we find an additional mode with $\mathrm{Im}\,\Omega > 0$. With our convention $e^{-i\omega t}$, this corresponds to exponential growth in time. The instability appears in the near-global-AdS regime, in the intermediate regime, and in the large black hole regime. A near-Dirichlet refinement shows that no finite critical angle is observed down to $\theta = \pi/64$. Within the numerical range probed, the instability appears as soon as the Dirichlet endpoint is deformed.

The paper is organized as follows. In Sec. II we introduce the Schwarzschild–AdS geometry, fix conventions, and derive the scalar radial equation. We also summarize the standard boundary conditions used for scalar, electromagnetic and gravitational perturbations in the SAdS literature. Section III describes the Chebyshev spectral method, the quadratic matrix-pencil formulation, the resolution-triplet matching procedure, and the modification needed to impose the generalized coefficient boundary condition. In Sec. IV, we present the numerical spectra. We first benchmark the Dirichlet scalar problem against existing literature and then study the instability induced by the generalized coefficient condition. Finally, Sec. V summarizes the results and discusses possible extensions to electromagnetic and gravitational perturbations.

II. METRIC AND EQUATIONS OF MOTION

To introduce the QNM eigenvalue problem and fix our conventions, we first review the minimally coupled massless scalar equation on a four dimensional SAdS background and then quote the analogous electromagnetic and gravitational master equations. We use units $c = G_N = \hbar = 1$. The line element is written as [6, 12]

$$ds^2 = -f(r)dt^2 + \frac{dr^2}{f(r)} + r^2\left(d\vartheta^2 + \sin^2\vartheta\, d\varphi^2\right), \qquad f(r) = 1 - \frac{2M}{r} + \frac{r^2}{R^2}, \tag{1}$$

where M is the black hole mass parameter and R is the AdS radius, related to the negative cosmological constant by $\Lambda = -3/R^2$. It is convenient to replace M by the event-horizon radius r_+. Since r_+ is the largest positive root of $f(r)$, the condition $f(r_+) = 0$ gives

$$2M = r_+\left(1 + \frac{r_+^2}{R^2}\right), \tag{2}$$

and hence

$$f(r) = 1 + \frac{r^2}{R^2} - \frac{r_+}{r}\left(1 + \frac{r_+^2}{R^2}\right). \tag{3}$$

We define the tortoise coordinate by

$$\frac{dr_*}{dr} = \frac{1}{f(r)}, \qquad \frac{d}{dr_*} = f(r)\frac{d}{dr}. \tag{4}$$

For a minimally coupled massless scalar field we use the separation

$$\phi(t, r, \vartheta, \varphi) = e^{-i\omega t}Y_{\ell m}(\vartheta, \varphi)\frac{\psi(r)}{r}. \tag{5}$$

The scalar equation takes the Schrödinger-type form [6, 12]

$$\frac{d^2\psi}{dr_*^2} + \left[\omega^2 - U(r)\right]\psi(r) = 0, \tag{6}$$

or equivalently

$$f(r)\frac{d}{dr}\left(f(r)\frac{d\psi}{dr}\right) + \left[\omega^2 - U(r)\right]\psi(r) = 0, \tag{7}$$

where

$$U(r) = f(r)\left[\frac{1}{r}\frac{df}{dr} + \frac{\ell(\ell+1)}{r^2}\right]. \tag{8}$$

The allowed multipoles for scalar perturbations are $\ell = 0, 1, 2, \ldots$. With the time dependence $e^{-i\omega t}$, decaying modes have $\Im\omega < 0$. At the event horizon, the QNM condition is purely ingoing behavior, i.e.

$$\psi(r) \sim e^{-i\omega r_*}, \qquad r \to r_+. \tag{9}$$

The boundary condition at the AdS boundary depends on the field and on the physical problem under consideration (see Table I for a general overview). For the standard minimally coupled massless scalar benchmark one imposes

$$\psi(r) \sim r^{-2}, \qquad r \to \infty. \tag{10}$$

More general scalar boundary conditions will be specified separately. Finally, we introduce the dimensionless variables $x = r/R$, $x_+ = r_+/R$, $\Omega = \omega R$, and $\mu = M/R$. From Eq. (3), one obtains

$$f(x) = 1 + x^2 - \frac{x_+(1 + x_+^2)}{x} = (x - x_+)\frac{x^2 + x_+x + x_+^2 + 1}{x}. \tag{11}$$

TABLE I: Representative boundary conditions used in Schwarzschild–AdS QNM calculations. We use the convention $e^{-i\omega t}$ and $dr_*/dr = 1/f(r)$. The horizon condition is purely ingoing in all cases; the main differences arise at the AdS boundary.

Spin	**Representative use in the literature**	**Horizon condition**	**AdS-boundary condition**
$s = 0$	Minimally coupled massless scalar; standard Horowitz–Hubeny-type benchmark [6–8, 11].	Purely ingoing: $\psi \sim e^{-i\omega r_*}$ as $r \to r_+$.	Dirichlet/normalizable falloff: $\Phi \to 0$. With $\Phi = \psi/r$, this gives $\psi \sim r^{-2}$ as $r \to \infty$.
$s = 0$	Scalar fields with admissible slow and fast falloffs, e.g. effective mass in the Breitenlohner–Freedman window [5, 16].	Purely ingoing, or regular after factoring out the ingoing horizon behavior.	Robin/mixed condition: $\Phi \sim z^{\Delta_-}\phi_1 + z^{\Delta_+}\phi_2$, with $\phi_2 = \kappa\phi_1$. Dirichlet and Neumann are limiting or special cases.
$s = 1$	Classic electromagnetic SAdS calculations [8, 11, 12].	Purely ingoing: $\psi \sim e^{-i\omega r_*}$.	Standard field-vanishing reflective condition: $\psi \to 0$, equivalently $\psi \sim r^{-1}$. If $\psi \sim a_0 + a_1/r + \cdots$, this corresponds to $a_0 = 0$.
$s = 1$	Maxwell perturbations with vanishing-energy-flux boundary conditions [13, 18].	Purely ingoing.	Two reflecting branches are possible: $a_0 = 0$ or $a_1 = 0$ in $\psi \sim a_0 + a_1/r + \cdots$. In Teukolsky variables these appear as two Robin boundary conditions.
$s = 2$	Older Regge–Wheeler/Zerilli SAdS calculations [8, 12].	Purely ingoing.	Often implemented as field-vanishing conditions for the master wavefunctions. This was the convention used in several older numerical tables.
$s = 2$	Fixed-boundary-metric or AdS/CFT-motivated gravitational perturbations [14, 15].	Purely ingoing.	Parity-dependent: axial/odd Dirichlet-type; polar/even Robin-type.

Once x_+ is specified, the dimensionless mass parameter is recovered from

$$2\mu = x_+(1 + x_+^2), \tag{12}$$

which is the dimensionless form of Eq. (2). Thus, after measuring all lengths in units of the AdS radius R, the geometry is completely characterized by the single dimensionless parameter $x_+ = r_+/R$. Fixing different values of x_+ is therefore equivalent to choosing different values of the dimensionless mass parameter $\mu =$ according to (12). Since $d\mu/dx_+ = (1 + 3x_+^2)/2 > 0$, the relation (12) is one-to-one for $x_+ > 0$. In what follows we label the spectra by x_+, which also allows a direct distinction between small ($x_+ \ll 1$), intermediate ($x_+ \sim 1$), and large ($x_+ \gg 1$) SAdS black holes. Finally, in the new variable x the radial equation (7) becomes

$$f(x)\frac{d}{dx}\left(f(x)\frac{d\psi}{dx}\right) + \left[\Omega^2 - U(x)\right]\psi(x) = 0, \quad \Omega = \omega R, \tag{13}$$

where

$$U(x) = f(x)\left[\frac{1}{x}\frac{df}{dx} + \frac{\ell(\ell+1)}{x^2}\right]. \tag{14}$$

By means of the substitution $z = x/x_+$, which sends $x_h \to 1$, Eq. (6) can be expressed in the equivalent form

$$f(z)\frac{d}{dz}\left(f(z)\frac{d\psi}{dz}\right) + \left[x_+^2\Omega^2 - V(z)\right]\psi(z) = 0\,, \qquad V(z) = f(z)\left[\frac{1}{z}\frac{df}{dz} + \frac{\ell(\ell+1)}{z^2}\right]\,, \qquad \Omega = \omega R\,, \tag{15}$$

with $f(z)$ given by

$$f(z) = (z-1)\frac{x_+^2 z^2 + x_+^2 z + x_+^2 + 1}{z}\,. \tag{16}$$

In the following analysis, we focus on computing the QNMs for the spectral problem stated in Eq. (15). For this purpose, we represent Ω as $\Omega = \Omega_R + i\Omega_I$, where $\Omega_I < 0$ ensures that the perturbation is damped in time.

1. The scalar case with Dirichlet boundary conditions at the conformal AdS boundary

The boundary conditions are set so that the radial field exhibits inward radiation at the event horizon and vanishes at the conformal AdS boundary. Moreover, to compute the QNMs using the spectral method, we must recast the differential equation in Eq. (15) and the appropriate boundary conditions over the compact interval $[-1,1]$. This adjustment is essential, as the method expands the regular part of the eigenfunctions by means of Chebyshev polynomials. To formalize the QNM boundary conditions at the event horizon and at the conformal AdS boundary, we first need to determine the asymptotic behavior of the radial solution ψ as $z \to 1^+$ and as $z \to +\infty$. We can then extract the QNM boundary conditions from this asymptotic data. We split our analysis by examining the behavior of the radial field in two different regions

1. *Asymptotic behavior as* $z \to 1^+$: First of all, we observe that $z = 1$ is a simple zero of $f(z)$ because $f(1) = 0$ and $f^{'}(1) = 1 + 3x_+^2 \neq 0$, where the prime symbol stands for differentiation with respect to z. Furthermore, it is convenient to reformulate Eq. (15) in the form

$$\frac{d^2\psi}{dz^2} + p(z)\frac{d\psi}{dz} + q(z)\psi(z) = 0\ , \quad p(z) = \frac{f^{'}(z)}{f(z)}\ , \quad q(z) = \frac{x_+^2\Omega^2 - V(z)}{f^2(z)}\ . \tag{17}$$

Since $p(z)$ and $q(z)$ have poles of order one and two at $z = 1$, respectively, this point is classified as a regular singular point of Eq. (17), according to Frobenius theory [19]. Hence, we can construct solutions of the form

$$\psi(z) = (z-1)^\rho \sum_{\kappa=0}^{\infty} a_\kappa (z-1)^\kappa\ . \tag{18}$$

The leading behavior at $z = 1$ is represented by the term $(z-1)^\rho$ where ρ is determined by the indicial equation,

$$\rho(\rho - 1) + \mathcal{P}_0\rho + \mathcal{Q}_0 = 0 \tag{19}$$

with

$$\mathcal{P}_0 = \lim_{z\to 1}(z-1)p(z) = 1\ , \qquad \mathcal{Q}_0 = \lim_{z\to 1}(z-1)^2 q(z) = \left(\frac{x_+\Omega}{f^{'}(1)}\right)^2\ . \tag{20}$$

The roots of Eq. (19) are $\rho_\pm = \pm i x_+\Omega/f^{'}(1)$ and the correct QNM boundary condition at $z = 1$ reads as

$$\psi \underset{z\to 1^+}{\longrightarrow} (z-1)^{-ix_+\alpha\Omega}, \qquad \alpha = \frac{1}{f^{'}(1)} = \frac{1}{3x_+^2 + 1}\ . \tag{21}$$

2. *Asymptotic behavior as* $z \to +\infty$: Unlike in asymptotically flat or de Sitter geometries, the AdS boundary is not a wave zone. Hence, no outgoing-wave condition is imposed at infinity. Instead, the radial solution has a power-law asymptotic behavior. From Eq. (16) we have

$$f(z) = x_+^2 z^2 + 1 - \frac{1 + x_+^2}{z}, \tag{22}$$

and therefore

$$p(z) = \frac{2}{z} + \mathcal{O}\left(\frac{1}{z^3}\right), \quad q(z) = -\frac{2}{z^2} + \mathcal{O}\left(\frac{1}{z^4}\right). \tag{23}$$

Thus, the point at infinity is a regular singular point [20]. Looking for solutions of the form $\psi(z) \sim z^\rho$ as $z \to +\infty$, one obtains the indicial equation $\rho^2 + \rho - 2 = 0$ whose roots are $\rho_1 = -2$, and $\rho_2 = 1$. The two asymptotic branches are therefore

$$\psi(z) \underset{z\to+\infty}{\longrightarrow} Az + \frac{B}{z^2}. \tag{24}$$

The standard vanishing field condition at the AdS boundary sets $A = 0$ [6, 11, 12], so that the QNM boundary condition at the conformal AdS boundary is

$$\psi(z) \underset{z\to+\infty}{\longrightarrow} \frac{B}{z^2}, \qquad z \to +\infty. \tag{25}$$

At this point, we transform the radial function $\psi(z)$ into a new radial function $\Phi(z)$ such that the QNM boundary conditions are automatically implemented and $\Phi(z)$ is regular at $z=1$ and at the conformal AdS boundary. To this end, we consider the transformation

$$\psi(z)=z^{-2+i\beta\Omega}(z-1)^{-i\beta\Omega}\Phi(z),\quad \beta=\frac{x_+}{3x_+^2+1}\,. \tag{26}$$

If we substitute Eq. (26) into Eq. (15), we end up with the following ordinary differential equation for the radial eigenfunctions

$$P_2(z)\Phi^{''}(z)+P_1(z)\Phi^{'}(z)+P_0(z)\Phi(z)=0\,, \tag{27}$$

with

$$P_2(z)=z^2(z-1)^2f^2(z)\,, \tag{28}$$
$$P_1(z)=(z-z^2)f(z)\left[(z-z^2)f^{'}(z)+2f(z)(i\beta\Omega+2z-2)\right]\,, \tag{29}$$
$$P_0(z)=-\Omega^2Q_2(z)+i\Omega Q_1(z)+Q_0(z)\,, \tag{30}$$
$$Q_2(z)=\beta^2f^2(z)-x_+^2z^2(z-1)^2,\quad Q_1(z)=\beta(z-z^2)f(z)f^{'}(z)+\beta f^2(z)(6z-5)\,, \tag{31}$$
$$Q_0(z)=-2z(z-1)^2f(z)f^{'}(z)+6f^2(z)(z-1)^2-(z-z^2)^2V(z). \tag{32}$$

Let us now introduce the transformation

$$z(y)=\frac{2}{1-y}\,, \tag{33}$$

mapping the event horizon to $y=-1$ and the conformal AdS boundary to $y=1$. In the following, a dot denotes differentiation with respect to the new variable y. Then, Eq. (27) becomes

$$S_2(y)\ddot{\Phi}(y)+S_1(y)\dot{\Phi}(y)+S_0(y)\Phi(y)=0\,, \tag{34}$$

where

$$S_2(y)=f^2(y)(1+y)^2\,, \tag{35}$$
$$S_1(y)=-2i\beta\Omega(1+y)f^2(y)-\frac{6(1+y)^2}{1-y}f(y)^2+(1+y)^2f(y)\dot{f}(y)\,, \tag{36}$$
$$S_0(y)=\Omega^2\Sigma_2(y)+i\Omega\Sigma_1(y)+\Sigma_0(y) \tag{37}$$

with

$$\Sigma_2(y)=\frac{4x_+^2(1+y)^2}{(1-y)^4}-\beta^2f^2(y),\quad \Sigma_1(y)=-\beta(1+y)f(y)\dot{f}(y)+\frac{\beta(7+5y)}{1-y}f^2(y)\,, \tag{38}$$
$$\Sigma_0(y)=-\frac{2(1+y)^2}{1-y}f(y)\dot{f}(y)+2\left(\frac{1+y}{1-y}\right)^2\left[3f^2(y)-\frac{2V(y)}{(1-y)^2}\right]\,. \tag{39}$$

Notice that we must also require that $\Phi(y)$ is regular at $y=\pm1$. As a result of the transformation introduced above, we have

$$f(y)=(1+y)g(y),\quad g(y)=\frac{(x_+^2+1)y^2-2y(2x_+^2+1)+7x_+^2+1}{2(1-y)^2}, \tag{40}$$
$$V(y)=\frac{1}{4}(1-y)^2f(y)\left[(1-y)\dot{f}(y)+\ell(\ell+1)\right]\,. \tag{41}$$

Table II shows that the coefficients of the differential equation (34) share a common zero of order 3 at $y=-1$ while $y=1$ is a pole of order 4 for $S_2(y)$ and order 5 for $S_1(y)$ and $S_0(y)$. Hence, in order to apply the spectral method, we need to multiply Eq. (34) by the factor $(1-y)^5/(1+y)^3$. As a result, we end up with the following differential equation

$$M_2(y)\ddot{\Phi}(y)+M_1(y)\dot{\Phi}(y)+M_0(y)\Phi(y)=0\,, \tag{42}$$

TABLE II: *Classification of the points $y = \pm 1$ for the relevant functions defined by Eqs (35)–(41). The abbreviations z ord n and p ord m stand for zero of order n and pole of order m, respectively.*

y	$f(y)$	$V(y)$	$S_2(y)$	$S_1(y)$	$S_0(y)$
-1	z ord 1	z ord 1	z ord 4	z ord 3	z ord 3
$+1$	p ord 2	p ord 2	p ord 4	p ord 5	p ord 5

where

$$M_2(y) = (1-y)^5(1+y)g^2(y)\ , \tag{43}$$
$$M_1(y) = i\Omega N_1(y) + N_0(y)\ , \qquad M_0(y) = \Omega^2 C_2(y) + i\Omega C_1(y) + C_0(y) \tag{44}$$

with

$$N_1(y) = -2\beta(1-y)^5 g^2(y)\ , \tag{45}$$
$$N_0(y) = -6(1-y)^4(1+y)g^2(y) + (1-y)^5 g(y)\left[(1+y)\dot{g}(y) + g(y)\right]\ , \tag{46}$$
$$C_2(y) = \frac{1-y}{1+y}\left[4x_+^2 - \beta^2 g^2(y)(1-y)^4\right]\ , \tag{47}$$
$$C_1(y) = \beta\frac{(1-y)^4}{1+y}\left\{(7+5y)g^2(y) - (1-y)g(y)\left[(1+y)\dot{g}(y) + g(y)\right]\right\}\ , \tag{48}$$
$$C_0(y) = -(1-y)^3 g(y)\left[3(1-y^2)\dot{g}(y) - 3(3y+1)g(y) + \ell(\ell+1)\right]\ . \tag{49}$$

It can be easily verified with Maple that

$$\lim_{y\to 1^-} M_2(y) = 0 = \lim_{y\to -1^+} M_2(y)\ , \tag{50}$$
$$\lim_{y\to 1^-} M_1(y) = -32x_+^4, \qquad \lim_{y\to -1^+} M_1(y) = iD_1\Omega + D_0\ , \tag{51}$$
$$\lim_{y\to 1^-} M_0(y) = \frac{16i\Omega x_+^5}{3x_+^2+1}, \qquad \lim_{y\to -1^+} M_0(y) = B_2\Omega^2 + iB_1\Omega + B_0\ , \tag{52}$$

where

$$D_1 = -16x_+(3x_+^2+1), \quad D_0 = 8(3x_+^2+1)^2\ , \tag{53}$$
$$B_2 = \frac{8x_+^2(3x_+^2+2)}{3x_+^2+1}, \quad B_1 = 12x_+(5x_+^2+2), \quad B_0 = -4(3x_+^2+1)\left[\ell(\ell+1) + 3(3x_+^2+1)\right]\ . \tag{54}$$

In the final step leading to the application of the spectral method, we recast the differential equation, Eq. (42), into the following form

$$L_0\left[\Phi,\dot{\Phi},\ddot{\Phi}\right] + iL_1\left[\Phi,\dot{\Phi},\ddot{\Phi}\right]\Omega + L_2\left[\Phi,\dot{\Phi},\ddot{\Phi}\right]\Omega^2 = 0 \tag{55}$$

with

$$L_0\left[\Phi,\dot{\Phi},\ddot{\Phi}\right] = L_{00}(y)\Phi + L_{01}(y)\dot{\Phi} + L_{02}(y)\ddot{\Phi}\ , \tag{56}$$
$$L_1\left[\Phi,\dot{\Phi},\ddot{\Phi}\right] = L_{10}(y)\Phi + L_{11}(y)\dot{\Phi} + L_{12}(y)\ddot{\Phi}\ , \tag{57}$$
$$L_2\left[\Phi,\dot{\Phi},\ddot{\Phi}\right] = L_{20}(y)\Phi + L_{21}(y)\dot{\Phi} + L_{22}(y)\ddot{\Phi}\ . \tag{58}$$

Moreover, in Table III, we have summarised the L_{ij} appearing in Eqs (56)–(58) and their limiting values at $y = \pm 1$.

TABLE III: *Definitions of the coefficients L_{ij} and their corresponding behaviors at the endpoints of the interval $-1 \leq y \leq 1$. The symbols appearing in this table have been defined in Eqs (45)-(49).*

(i,j)	$\lim_{y\to -1^+} L_{ij}$	L_{ij}	$\lim_{y\to 1^-} L_{ij}$
$(0,0)$	B_0	C_0	0
$(0,1)$	D_0	N_0	$-32x_+^4$
$(0,2)$	0	M_2	0
$(1,0)$	B_1	C_1	$16x_+^5/(3x_+^2+1)$
$(1,1)$	D_1	N_1	0
$(1,2)$	0	0	0
$(2,0)$	B_2	C_2	0
$(2,1)$	0	0	0
$(2,2)$	0	0	0

2. *The scalar case with generalized coefficient boundary conditions at the conformal AdS boundary*

The previous subsection considered the standard vanishing-field condition at the conformal AdS boundary, corresponding to the removal of the slow asymptotic branch of the minimally coupled massless scalar field. In the present subsection, we consider instead a one-parameter family of generalized coefficient boundary conditions relating the two asymptotic coefficients of the same massless scalar field. We stress that, for the minimally coupled massless scalar, this should be viewed as a deformation of the spectral problem rather than as the usual alternative quantization available for scalars in the Breitenlohner–Freedman window [5, 16]. In particular, the standard finite-energy prescription selects the Dirichlet branch, whereas the mixed conditions introduced below are imposed as generalized boundary conditions on the QNM eigenvalue problem [17]. As before, the boundary condition at the event horizon is purely ingoing. The main difference with respect to the Dirichlet case is that we shall not remove the slow branch at the AdS boundary from the beginning. Instead, the relation between the slow and fast coefficients will be imposed as a spectral boundary condition. We again start from Eq. (15), and we determine the asymptotic behavior of the radial function ψ in the two relevant regions. Concerning the case $z \to 1^+$, the analysis remains unchanged and the ingoing QNM condition is therefore given by (21). As $z \to +\infty$, we have two independent asymptotic branches captured by (24). The standard Dirichlet condition sets $A = 0$. In the present generalized problem, we instead impose a linear relation between A and B. To express this relation in terms of the scalar field itself, we use (24) together with the rescalings $z = r/r_+$, and $x_+ = r_+/R$. Then, a straightforward computation gives

$$\frac{\psi(z)}{z} \underset{z\to+\infty}{\longrightarrow} A + \frac{B}{z^3}. \tag{59}$$

Thus, the two independent coefficients of the scalar field may be identified with A and B. We impose the one-parameter family of generalized coefficient boundary conditions

$$A\cos\theta + B\sin\theta = 0, \quad 0 \leq \theta\frac{\pi}{2}. \tag{60}$$

The endpoint $\theta = 0$ gives the standard Dirichlet condition $A = 0$, while $\theta = \pi/2$ gives the coefficient-Neumann condition $B = 0$. We shall refer to intermediate values of θ as generalized coefficient Robin boundary conditions. These conditions should not be confused with the standard Robin family for scalars in the alternative-quantization window. In order to construct an ansatz which takes into account the boundary condition (60) together with the correct asymptotic behavior at the event horizon and at the AdS boundary, we consider the following educated guess

$$\psi(z) = z\,\mathcal{H}(z;\Omega)\,\chi(z), \tag{61}$$

where

$$\mathcal{H}(z;\Omega) = e^{-ix_+\alpha\Omega\left[\ln\left(1-\frac{1}{z}\right)+\frac{1}{z}+\frac{1}{2z^2}+\frac{1}{3z^3}\right]}. \tag{62}$$

Furthermore, we require that $\chi(Z)$ is regular at the event horizon and asymptotically away from it. The factor $H(z;\Omega)$ has two useful properties. Near the event horizon, $w \to 1$, it behaves as

$$\mathcal{H}(z;\Omega) \underset{z\to 1^+}{\longrightarrow} (z-1)^{-ix_+\alpha\Omega}, \tag{63}$$

and hence Eq. (61) implements the ingoing QNM behavior. Near the conformal AdS boundary, $z \to +\infty$, one has

$$\mathcal{H}(z;\Omega) = 1 + \mathcal{O}\left(\frac{1}{z^4}\right). \tag{64}$$

We therefore write the regular function $\chi(z)$ as

$$\chi(z) = A + \frac{\mathcal{C}_2}{z^2} + \frac{B}{z^3} + \mathcal{O}\left(\frac{1}{z^4}\right), \qquad z \to +\infty. \tag{65}$$

The coefficient of z^{-1} vanishes, while C_2 is fixed by the differential equation. For the massless scalar one finds

$$\mathcal{C}_2 = \frac{\Omega^2 - \ell(\ell+1)}{2x_+^2} A. \tag{66}$$

This implies that

$$\frac{\psi(z)}{z} = A + \frac{\mathcal{C}_2}{z^2} + \frac{B}{z^3} + \mathcal{O}\left(\frac{1}{z^4}\right). \tag{67}$$

and the coefficients A and B in Eq. (60) are directly encoded in the expansion of $\chi(z)$. Substitution of Eq. (61) into Eq. (15) gives

$$\widetilde{P}_2(z)\chi''(z) + \widetilde{P}_1(z)\chi'(z) + \widetilde{P}_0(z)\chi(z) = 0, \tag{68}$$

where primes now denote differentiation with respect to z, and

$$\widetilde{P}_2(z) = z^8(z-1)^2 f^2(z)\,, \tag{69}$$

$$\widetilde{P}_1(z) = z^4(z-1)f(z)\left[z^4(z-1)f'(z) + 2f(z)(z^4 - z^3 - i\beta\Omega)\right]\,, \tag{70}$$

$$\widetilde{P}_0(z) = -\Omega^2\widetilde{Q}_2(z) + i\Omega\widetilde{Q}_1(z) + \widetilde{Q}_0(z)\,, \tag{71}$$

$$\widetilde{Q}_2(z) = \beta^2 f^2(z) - x_+^2 z^8(z-1)^2, \quad \widetilde{Q}_1(z) = -\beta z^4(z-1)f(z)f'(z) + \beta z^3 f(z)^2(3z-2)\,, \tag{72}$$

$$\widetilde{Q}_0(z) = z^7(z-1)^2 f(z)f'(z) - z^8(z-1)^2 V(z). \tag{73}$$

We compactify the exterior region using the same transformation (33). The event horizon is located at $y = -1$, while the conformal AdS boundary is located at $y = 1$. Moreover, by a dot we denote differentiation with respect to y. Then, Eq. (68) becomes

$$\widetilde{S}_2(y)\ddot{\chi}(y) + \widetilde{S}_1(y)\dot{\chi}(y) + \widetilde{S}_0(y)\chi(y) = 0, \tag{74}$$

where

$$\widetilde{S}_2(y) = \frac{64 f^2(y)(1+y)^2}{(1-y)^6}\,, \tag{75}$$

$$\widetilde{S}_1(y) = -\frac{128 f^2(y)(1+y)^2}{(1-y)^7} + \frac{8(1+y)f(y)}{(1-y)^7}\left\{8(1-y^2)\dot{f}(y) + 2f(y)\left[16 - 8(1-y) - i\beta\Omega(1-y)^4\right]\right\}\,, \tag{76}$$

$$\widetilde{S}_0(y) = \Omega^2\widetilde{\Sigma}_2(y) + i\Omega\widetilde{\Sigma}_1(y) + \widetilde{\Sigma}_0(y) \tag{77}$$

with

$$\widetilde{\Sigma}_2(y) = \frac{256(1+y)^2 x_+^2}{(1-y)^1 0} - \beta^2 f^2(y), \quad \widetilde{\Sigma}_1(y) = -\frac{8\beta(1+y)}{(1-y)^3} f(y)\dot{f}(y) + \frac{16\beta(2+y)f^2(y)}{(1-y)^4}\,, \tag{78}$$

$$\widetilde{\Sigma}_0(y) = \frac{64(1+y)^2}{(1-y)^7} f(y)\dot{f}(y) - \frac{256(1+y)^2}{(1-y)^{10}} V(y)\,. \tag{79}$$

TABLE IV: *Classification of the points $y = \pm 1$ for the relevant functions defined by Eqs (75)–(79). The abbreviations z ord n and p ord m stand for zero of order n and pole of order m, respectively.*

y	$f(y)$	$V(y)$	$S_2(y)$	$S_1(y)$	$S_0(y)$
-1	z ord 1	z ord 1	z ord 4	z ord 3	z ord 3
$+1$	p ord 2	p ord 2	p ord 10	p ord 11	p ord 10

In practice, Eq. (74) is imposed at the interior collocation points. Furthermore, $f(y)$ and $V(y)$ are given as in (40) and (41), respectively.

Table IV shows that the coefficients of the differential equation (74) share a common zero of order 3 at $y = -1$ while $y = 1$ is a pole of order 11 for $\widetilde{S}_1(y)$ and order 10 for $\widetilde{S}_2(y)$ and $\widetilde{S}_0(y)$. Hence, in order to apply the spectral method, we need to multiply Eq. (74) by the factor $(1-y)^{11}/(1+y)^3$. As a result, we end up with the following differential equation

$$\widetilde{M}_2(y)\ddot{\chi}(y) + \widetilde{M}_1(y)\dot{\chi}(y) + \widetilde{M}_0(y)\chi(y) = 0 \,, \tag{80}$$

where

$$\widetilde{M}_2(y) = 64(1-y)^5(1+y)g^2(y) \,, \tag{81}$$

$$M_1(y) = i\Omega N_1(y) + N_0(y) \,, \qquad M_0(y) = \Omega^2 C_2(y) + i\Omega C_1(y) + C_0(y) \tag{82}$$

with

$$\widetilde{N}_1(y) = -16\beta(1-y)^8 g^2(y) \,, \quad \widetilde{N}_0(y) = 64(1-y)^5 g(y)\left[g(y) + (1+y)\dot{g}(y)\right] \,, \tag{83}$$

$$\widetilde{C}_2(y) = \frac{1-y}{1+y}\left[256x_+^2 - \beta^2 g^2(y)(1-y)^{10}\right] \,, \tag{84}$$

$$\widetilde{C}_1(y) = 8\beta g(y)(1-y)^7\left[3g(y) - (1-y)\dot{g}(y)\right] \,, \quad \widetilde{C}_0(y) = -64\ell(\ell+1)g(y)(1-y)^3 \,. \tag{85}$$

It can be easily verified with Maple that

$$\lim_{y\to 1^-} \widetilde{M}_2(y) = 0 = \lim_{y\to -1^+} \widetilde{M}_2(y) \,, \tag{86}$$

$$\lim_{y\to 1^-} \widetilde{M}_1(y) = 1024x_+^4, \qquad \lim_{y\to -1^+} \widetilde{M}_1(y) = i\widetilde{D}_1\Omega + \widetilde{D}_0 \,, \tag{87}$$

$$\lim_{y\to 1^-} \widetilde{M}_0(y) = 0, \qquad \lim_{y\to -1^+} \widetilde{M}_0(y) = \widetilde{B}_2\Omega^2 + i\widetilde{B}_1\Omega + \widetilde{B}_0 \,, \tag{88}$$

where

$$\widetilde{D}_1 = -1024x_+(3x_+^2+1), \quad \widetilde{D}_0 = 512(3x_+^2+1)^2 \,, \tag{89}$$

$$\widetilde{B}_2 = \frac{512x_+^2(12x_+^2+5)}{3x_+^2+1}, \quad \widetilde{B}_1 = 768x_+(2x_+^2+1), \quad \widetilde{B}_0 = -256\ell(\ell+1)(3x_+^2+1) \,. \tag{90}$$

In the final step leading to the application of the spectral method, we recast the differential equation, Eq. (80), into the following form

$$\widetilde{L}_0\left[\chi,\dot{\chi},\ddot{\chi}\right] + i\widetilde{L}_1\left[\chi,\dot{\chi},\ddot{\chi}\right]\Omega + \widetilde{L}_2\left[\chi,\dot{\chi},\ddot{\chi}\right]\Omega^2 = 0 \tag{91}$$

with

$$\widetilde{L}_0\left[\chi,\dot{\chi},\ddot{\chi}\right] = \widetilde{L}_{00}(y)\chi + \widetilde{L}_{01}(y)\dot{\chi} + \widetilde{L}_{02}(y)\ddot{\chi} \,, \tag{92}$$

$$\widetilde{L}_1\left[\chi,\dot{\chi},\ddot{\chi}\right] = \widetilde{L}_{10}(y)\chi + \widetilde{L}_{11}(y)\dot{\chi} + \widetilde{L}_{12}(y)\ddot{\chi} \,, \tag{93}$$

$$\widetilde{L}_2\left[\chi,\dot{\chi},\ddot{\chi}\right] = \widetilde{L}_{20}(y)\chi + \widetilde{L}_{21}(y)\dot{\chi} + \widetilde{L}_{22}(y)\ddot{\chi} \,. \tag{94}$$

TABLE V: *Definitions of the coefficients $\widetilde{L}_{ij}$ and their corresponding behaviors at the endpoints of the interval $-1 \leq y \leq 1$. The symbols appearing in this table have been defined in Eqs (83)-(85).*

(i,j)	$\lim_{y\to -1^+} L_{ij}$	L_{ij}	$\lim_{y\to 1^-} L_{ij}$
$(0,0)$	$\widetilde{B}_0$	$\widetilde{C}_0$	0
$(0,1)$	$\widetilde{D}_0$	$\widetilde{N}_0$	$1024x_+^4$
$(0,2)$	0	$\widetilde{M}_2$	0
$(1,0)$	$\widetilde{B}_1$	$\widetilde{C}_1$	0
$(1,1)$	$\widetilde{D}_1$	$\widetilde{N}_1$	0
$(1,2)$	0	0	0
$(2,0)$	$\widetilde{B}_2$	$\widetilde{C}_2$	0
$(2,1)$	0	0	0
$(2,2)$	0	0	0

Moreover, in Table V, we have summarised the $\widetilde{L}_{ij}$ appearing in Eqs (92)–(94) and their limiting values at $y = \pm 1$.

Furthermore, the endpoint $y = 1$ is used to impose the generalized boundary condition (60). To express Eq. (60) in terms of $\chi(y)$, we use (33) together with (65). A straightforward computation yields

$$A = \chi(1), \qquad B = -\frac{4}{3}\chi^{(3)}(1), \tag{95}$$

where the third derivative is with respect to y. The generalized coefficient boundary condition (60) becomes

$$\cos\theta\,\chi(1) - \frac{4}{3}\sin\theta\,\chi^{(3)}(1) = 0. \tag{96}$$

Thus, $\theta = 0$ implies $\chi(1) = 0$, which reproduces the standard Dirichlet condition, while $\theta = \pi/2$ requires $\chi^{(3)}(1) = 0$, which corresponds to the coefficient-Neumann condition. Finally, we expand the regular function $\chi(y)$ in Chebyshev polynomials,

$$\chi(y) = \sum_{j=0}^{N-1} b_j T_j(y). \tag{97}$$

Using the following identities in [21]

$$T_j(1) = 1, \qquad T_j^{(3)}(1) = \frac{j^2(j^2-1)(j^2-4)}{15}, \tag{98}$$

the boundary condition (96) gives the spectral boundary row

$$\sum_{j=0}^{N-1}\left[\cos\theta - \frac{4j^2(j^2-1)(j^2-4)}{45}\sin\theta\right] b_j = 0. \tag{99}$$

The remaining $N-1$ rows are obtained by imposing Eq. (74) at the interior collocation points. Since the differential-equation rows are quadratic in Ω, while the boundary row (99) is independent of Ω, the final discretized problem is again quadratic in Ω. Finally, the Dirichlet endpoint $\theta = 0$ provides a direct consistency check, because it must reproduce the standard scalar spectrum computed in the previous subsection. For $\theta \neq 0$, the spectrum describes a generalized coefficient deformation of the massless scalar QNM problem.

III. SPECTRAL METHOD AND EXTRACTION OF THE QNM SPECTRUM

In this section, we describe the numerical implementation used to compute the scalar QNM spectra discussed below. The method is based on a Chebyshev collocation discretization of the regularized radial equations derived

in the previous section. In all cases, the final algebraic problem is a quadratic matrix pencil in the dimensionless frequency $\Omega = \omega R$, namely

$$\left(\mathcal{M}_0 + i\Omega\mathcal{M}_1 + \Omega^2\mathcal{M}_2\right)\mathbf{b} = 0, \tag{100}$$

where $\mathbf{b} = (b_0, \ldots, b_{N-1})^T$ contains the Chebyshev expansion coefficients of the regular part of the radial eigenfunction. After compactifying the exterior region according to (33) the event horizon is located at $y = -1$, while the conformal AdS boundary is situated at $y = 1$. The regular radial function is expanded as

$$F_N(y) = \sum_{j=0}^{N-1} b_j T_j(y), \tag{101}$$

where $T_j(y)$ denotes the Chebyshev polynomial of degree j. The radial equation is then imposed at Chebyshev–Gauss collocation points. For a problem in which both boundary conditions are already built into the ansatz, we use N collocation points,

$$y_i = \cos\left(\frac{(2i-1)\pi}{2N}\right), \qquad i = 1, \ldots, N. \tag{102}$$

At each point y_i, the equation has the schematic form

$$L_0[F_N, \dot{F}_N, \ddot{F}_N](y_i) + i\Omega L_1[F_N, \dot{F}_N, \ddot{F}_N](y_i) + \Omega^2 L_2[F_N, \dot{F}_N, \ddot{F}_N](y_i) = 0, \tag{103}$$

where the dot denotes differentiation with respect to y. Since $F_N, \dot{F}_N, \ddot{F}_N$ are linear in the coefficients b_j, Eq. (103) generates one row of the matrices $\mathcal{M}_0, \mathcal{M}_1, \mathcal{M}_2$. Repeating this at all collocation points gives the quadratic pencil (100). For the standard massless scalar problem, the QNM boundary conditions are purely ingoing behavior at the event horizon and the vanishing-field condition at the conformal AdS boundary. As shown above, these conditions are implemented by means of the ansatz (26). Notice that the prefactor in Eq. (26) contains the ingoing horizon behavior and the standard scalar falloff $\psi \sim z^{-2}$ at the AdS boundary. Therefore, the remaining function Φ is regular at both endpoints. After the transformation (33), we expand

$$\Phi_N(y) = \sum_{j=0}^{N-1} b_j T_j(y), \tag{104}$$

and impose the regularized equation at the N points (102). This gives an $N \times N$ quadratic pencil of the form (100). For the generalized coefficient boundary condition, the AdS-boundary condition is not built into the ansatz. Instead, we factor out only the ingoing behavior at the horizon, while keeping both independent boundary coefficients available. The ansatz is provided by (61) together with (62) while the regular function $\chi(z)$ has the expansion (65) where the coefficients A and B are the two independent boundary data, which satisfy the constraint (60). Therein the endpoint $\theta = 0$ gives the usual Dirichlet condition $A = 0$, while $\theta = \pi/2$ gives the coefficient-Neumann condition $B = 0$. In the previous section, we showed that such a constraint can be expressed in terms of $\chi(y)$ as in (96). For the Chebyshev expansion (97), this boundary condition is represented by the spectral row (99). The generalized-boundary calculation is assembled as follows. We impose the regularized differential equation at $N-1$ Chebyshev–Gauss points

$$y_i = \cos\left(\frac{(2i-1)\pi}{2(N-1)}\right), \qquad i = 1, \ldots, N-1, \tag{105}$$

and use Eq. (99) as the final row. Hence, the matrices again have size $N \times N$, but their last row is a boundary-condition row rather than a collocation row. Since Eq. (99) is independent of Ω, the last row contributes only to $\mathcal{M}_0$. More precisely, we have

$$(\mathcal{M}_0)_{N,j+1} = \cos\theta - \frac{4}{45}\sin\theta\, j^2(j^2-1)(j^2-4), \qquad (\mathcal{M}_1)_{N,j+1} = (\mathcal{M}_2)_{N,j+1} = 0, \tag{106}$$

with $j \in \{0, \ldots, N-1\}$. In this formulation the leading matrix $\mathcal{M}_2$ is singular because the boundary row is independent of Ω. For the numerical solution, we therefore eliminate the boundary constraint before linearizing the quadratic pencil. To this end, let $c^T\mathbf{b} = 0$ denote the discrete boundary condition (99), and let Z be a basis of the nullspace of c^T. We write $\mathbf{b} = Z\mathbf{u}$. The first $N-1$ rows are the differential equation rows. If $\mathcal{R}$ denotes the matrix that selects these rows, the reduced quadratic pencil is

$$\left(\widehat{\mathcal{M}_0} + i\Omega\widehat{\mathcal{M}_1} + \Omega^2\widehat{\mathcal{M}_2}\right)\mathbf{u} = 0, \qquad \widehat{\mathcal{M}_k} = \mathcal{R}\mathcal{M}_k Z, \qquad k \in \{0, 1, 2\}. \tag{107}$$

This reduced pencil has dimension $(N-1)\times(N-1)$ and contains the boundary condition exactly at the discrete level. Since the problem reduces to a pencil of the form (100) for both boundary conditions, we use the companion linearization

$$\begin{pmatrix} 0 & I \\ -\mathcal{M}_0 & -i\mathcal{M}_1 \end{pmatrix} \begin{pmatrix} \mathbf{b} \\ \Omega\mathbf{b} \end{pmatrix} = \Omega \begin{pmatrix} I & 0 \\ 0 & \mathcal{M}_2 \end{pmatrix} \begin{pmatrix} \mathbf{b} \\ \Omega\mathbf{b} \end{pmatrix}. \tag{108}$$

For the generalized-boundary problem, the same linearization is applied to the reduced matrices $\widehat{\mathcal{M}_k}$ in Eq. (107). All matrix entries are exported from Maple with double digits precision, and the eigenvalue problem is solved in Julia using arbitrary-precision arithmetic. In the runs reported in the next section, we used 200 decimal digits for the matrix data and 220 decimal digits for the working precision. since the time dependence is $e^{-i\omega t}$, $\mathrm{Im}\,\Omega < 0$ corresponds to damped modes, while $\mathrm{Im}\,\Omega > 0$ corresponds to unstable modes. We do not discard modes with positive imaginary part. Instead, they are kept in the candidate set and classified separately. The classification tolerance is $\tau_{\mathrm{damp}} = 10^{-18}$. Modes with $|\Omega| < 10^{-30}$ are treated as static zero modes and are not counted as QNMs. Purely imaginary modes are identified by the criterion $|\,\mathrm{Re}\,\Omega| < 10^{-10}$. Moreover, for each choice of x_+, ℓ, and boundary parameter θ, we compute the spectrum at three spectral resolutions. In the production runs reported in this work, we use $N_1 = 190$, $N_2 = 195$, and $N_3 = 200$. Let $\Omega^{(1)}$, $\Omega^{(2)}$, and $\Omega^{(3)}$ denote matched modes at the three resolutions. The matching is performed by a nearest-neighbour search in the complex Ω-plane. Starting from a candidate mode at N_1, we find the closest unused candidate at N_2, and then the closest unused candidate at N_3. We define

$$d_{12} = \left|\Omega^{(1)} - \Omega^{(2)}\right|, \qquad d_{23} = \left|\Omega^{(2)} - \Omega^{(3)}\right|, \qquad d_{13} = \left|\Omega^{(1)} - \Omega^{(3)}\right|. \tag{109}$$

The mode is accepted if

$$\max(d_{12}, d_{23}, d_{13}) \le \max\left[\tau_{\mathrm{abs}}, \tau_{\mathrm{rel}} \max\left(1, |\Omega^{(1)}|, |\Omega^{(2)}|, |\Omega^{(3)}|\right)\right], \tag{110}$$

with $\tau_{\mathrm{abs}} = 10^{-4}$, and $\tau_{\mathrm{rel}} = 10^{-4}$. The quoted value of the frequency is the arithmetic average

$$\Omega_{\mathrm{avg}} = \frac{\Omega^{(1)} + \Omega^{(2)} + \Omega^{(3)}}{3}, \tag{111}$$

and the error estimator reported in the output files is

$$\Delta\Omega_{\mathrm{max}} = \max(d_{12}, d_{23}, d_{13}). \tag{112}$$

Only modes that pass the triplet criterion (110) are used in the final tables and plots. When displaying symmetric oscillatory pairs, we usually list only the branch with positive real part, since the radial equation has real coefficients and the companion branch is obtained from $\Omega \mapsto -\Omega^*$.

IV. NUMERICAL RESULTS FOR THE SADS

We now present the numerical spectra obtained with the Chebyshev spectral method described above. The discussion is divided into two parts. First, we study the standard Dirichlet boundary condition at the conformal AdS boundary, which provides a direct benchmark against the existing Schwarzschild–AdS scalar QNM literature. Second, we investigate the generalized coefficient condition $A\cos\theta + B\sin\theta = 0$, where the Dirichlet problem is recovered at $\theta = 0$, and analyze the boundary-condition-induced instability that appears for nonzero θ.

A. The scalar sector with Dirichlet boundary condition at the conformal AdS boundary

We begin with the standard massless scalar problem, imposing purely ingoing behavior at the event horizon and the Dirichlet, or vanishing-field, condition at the conformal AdS boundary. This case provides the natural benchmark for the spectral method, since it has been studied extensively with independent numerical techniques. Throughout this subsection we display frequencies in units of $\Omega = \omega R$, with the convention $\Omega = \Omega_R + i\Omega_I$, so that damped modes have $\Omega_I < 0$. Unless otherwise stated, only the branch with positive real part is tabulated, the negative-real branch being obtained from the symmetry $\Omega \mapsto -\Omega^*$. Table VI provides a first benchmark of our implementation for the fundamental scalar mode with $\ell = 0$. The agreement with the Horowitz–Hubeny power-series results is excellent over

the full range of black hole sizes considered. In particular, our values reproduce the quoted frequencies of [6] to all displayed digits, from the intermediate regime $x_+ \sim 1$ up to the large black hole regime $x_+ \gg 1$. This confirms that the compactification, the factorization of the horizon and AdS-boundary asymptotics, and the resulting quadratic matrix pencil reproduce the standard scalar SAdS QNM problem. The comparison with the interpolation matrix method of [10] is also very good for $x_+ \geq 0.6$. The only visible discrepancy occurs at $x_+ = 0.4$, where the value reported in [10] differs from both the Horowitz–Hubeny result and our spectral result. Since our value agrees with [6] at this point, we interpret this isolated difference as likely associated with the finite interpolation accuracy or with the tabulated value in [10], rather than with a change in the boundary-value problem. Apart from this point, the three methods give mutually consistent results. The table also illustrates the expected large-black hole scaling, that is for $x_+ = 50$ and $x_+ = 100$, the ratio Ω/x_+ is already approximately constant, $\Omega/x_+ \simeq 1.85 - 2.66i$, showing the onset of the linear scaling of scalar SAdS QNMs with the horizon radius. A more stringent comparison is given in Table VII, where we list a long sequence of overtones for the intermediate-size black hole $x_+ = 1$ with $\ell = 0$. The agreement with the data of [8] and with the asymptotic iteration method of [9] is again excellent. For the modes $n \in \{0, 1, 2\}$ and $n \geq 4$, our spectral values reproduce the quoted frequencies to all displayed digits and agree with the AIM values wherever those are available. The only apparent exception concerns the real part of the $n = 3$ entry in the data quoted in [8]. This mismatch was already noted in [9], where the corrected value was obtained independently using the Mathematica notebook associated with the QNM data of [2]. The AIM value $\Omega_{n=3} = 8.68233 - 9.74854i$ agrees with our spectral result, $\Omega_{n=3} = 8.682227 - 9.748517i$. Thus, the discrepancy should be interpreted as a previously identified tabulation issue rather than as a disagreement between methods. Table VII also shows that the overtones rapidly approach an almost uniformly spaced pattern. Using the last entries displayed in the table, one finds $\Omega_{20} - \Omega_{19} \simeq 1.96849 - 2.35043i$, which is close to the asymptotic spacing reported for intermediate-size SAdS black holes. This behavior is the finite-x_+ analogue of the nearly linear overtone structure that becomes more pronounced in the large black hole regime [8]. Table VIII extends the overtone comparison to two opposite regimes of the SAdS geometry. The case $x_+ = 0.2$ probes the small black hole regime, whereas $x_+ = 100$ is already deep in the large-black hole regime. In both cases, the agreement with [8] is excellent. For $x_+ = 100$, our spectral method reproduces all tabulated frequencies up to the displayed precision for $n \in \{0, \ldots, 10\}$. For $x_+ = 0.2$, the agreement is likewise excellent for the fundamental mode and for the overtones $n \in \{2, \ldots, 11\}$. The only visible exception is the imaginary part of the $n = 1$ entry. [8] quotes $\Omega = 4.07086 - 0.98966i$, whereas our spectral result is $\Omega = 4.070860 - 0.986629i$. This value is stable under changes of the spectral resolution, and the neighbouring overtones agree with the published data to the quoted accuracy. We therefore regard this as an isolated tabulation or rounding discrepancy in the literature value rather than as a failure of the spectral method. The two halves of Table VIII also illustrate the qualitative difference between small and large SAdS black holes. For $x_+ = 0.2$, the real parts are close to the pure-AdS scalar normal-mode sequence, while the imaginary parts remain comparatively small. This agrees with the expected small black hole behavior, i.e. as $x_+ \to 0$, the scalar QNM frequencies approach the pure-AdS normal modes $\Omega_{\rm AdS} = 2n + \ell + 3$, with the damping rate tending to zero. For $\ell = 0$, the fundamental pure-AdS value is $\Omega = 3$, and the $x_+ = 0.2$ fundamental mode, $\Omega = 2.475112 - 0.389925i$, is already moving toward this limit. By contrast, for $x_+ = 100$, both the real and imaginary parts grow approximately linearly with the overtone number. This is consistent with the large black hole asymptotic behavior found in [8],

$$\frac{\Omega_n}{x_+} \simeq (1.299 - 2.25i)n + 1.856 - 2.673i, \qquad n \to \infty, \tag{113}$$

or, equivalently,

$$\frac{\Omega_{n+1} - \Omega_n}{x_+} \simeq 1.299 - 2.25i. \tag{114}$$

For example, using the last two modes displayed for $x_+ = 100$, we find

$$\frac{\Omega_{20} - \Omega_{19}}{100} \simeq 1.29914 - 2.25002i, \tag{115}$$

which is already extremely close to the predicted asymptotic spacing. This agreement also demonstrates that the spectral method remains accurate for a long overtone sequence. As a final independent benchmark in the large black hole regime, Table IX compares our scalar spectrum with the continued-fraction calculation of [11]. This comparison is useful because their method is algorithmically different from both the Horowitz–Hubeny power-series approach and the present Chebyshev spectral discretization. The agreement is again at the level of the displayed digits for the low overtones, providing an additional validation of the implementation in the large black hole regime. We next focus on the small black hole regime in Table X. In this limit, $x_+ \ll 1$, the SAdS scalar spectrum approaches the normal mode spectrum of pure global AdS. The second column reports the fundamental $\ell = 0$ scalar frequencies computed by [7],

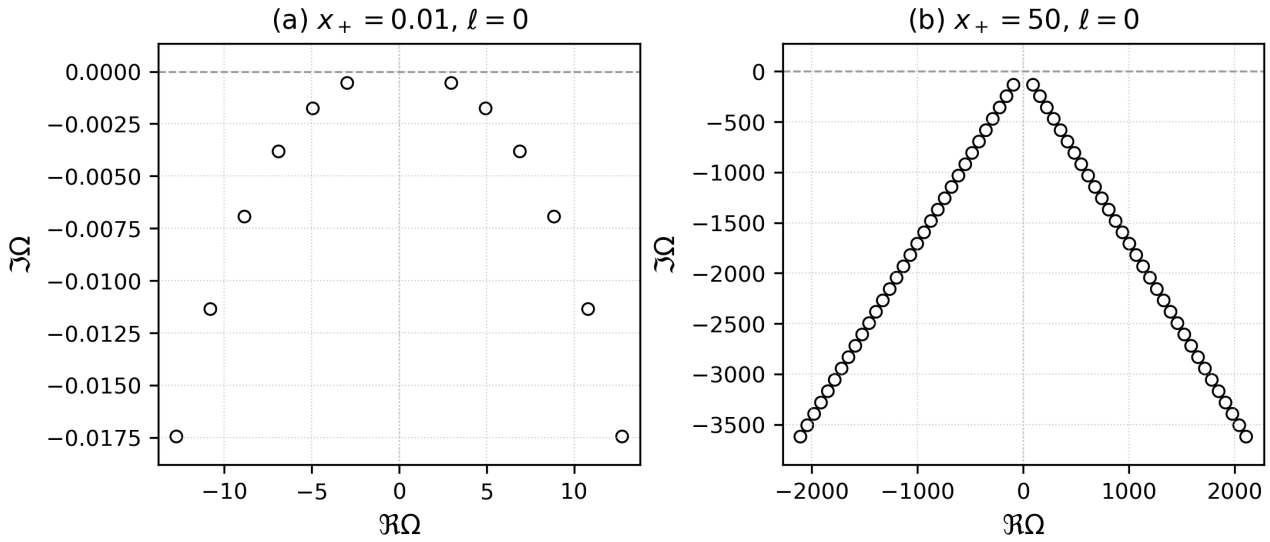


FIG. 1: *QNM spectrum of the massless scalar sector with $\ell = 0$ and the standard vanishing-field boundary condition at the conformal AdS boundary. The left panel shows the small black hole case $x_+ = 0.01$, while the right panel shows the large black hole case $x_+ = 50$. For $x_+ = 0.01$, the modes lie very close to the real axis and approach the pure-AdS normal-mode values $\Omega_{\rm AdS} = 2n + \ell + 3$. For $x_+ = 50$, the modes organize into two approximately straight, symmetry-related branches, reflecting the nearly uniform high-overtone spacing of large SAdS black holes.*

while the remaining columns show our spectral results for the fundamental mode and for the first three overtones. For the fundamental mode, the agreement with [7]'s data is very good throughout the range of x_+. This provides a nontrivial check of the spectral method in a regime where the damping rates become small and the numerical problem is more delicate. The approach to the pure AdS limit is transparent in the same table. For a massless scalar field in four-dimensional global AdS, the normal-mode frequencies are $\Omega_{\rm AdS} = 2n + \ell + 3$. Thus, for $\ell = 0$, the first four pure-AdS values are 3, 5, 7, 9. As x_+ decreases, the real parts of our QNM frequencies move toward precisely these values. For example, at $x_+ = 1/100$ we find $\Omega_0 \simeq 2.97379 - 0.00055i$, $\Omega_1 \simeq 4.93781 - 0.00176i$, $\Omega_2 \simeq 6.89526 - 0.00382i$, and $\Omega_3 \simeq 8.84749 - 0.00694i$. These are small dissipative deformations of the pure-AdS values 3, 5, 7, and 9, confirming that the Schwarzschild–AdS QNMs continuously approach the normal modes of empty AdS as the horizon radius tends to zero. The imaginary parts show the complementary trend, that is the damping decreases rapidly as x_+ is reduced. This is physically expected, since in the limit $x_+ \to 0$ the absorbing horizon shrinks away and the system approaches global AdS with reflecting boundary conditions. The small but nonzero negative imaginary parts in the last row therefore measure the weak absorption by a very small black hole. For the fundamental mode, this agrees with the conclusion in [7] that $\Omega_I \to 0^-$ as $x_+ \to 0$, while Ω_R tends to the corresponding AdS normal mode frequency. The additional columns $n \in \{1, 2, 3\}$ extend the comparison beyond the fundamental mode tabulated in [7]. They show that the same limiting behavior is not restricted to the fundamental frequency, since the first overtones also approach the expected AdS sequence $\Omega_{\rm AdS} = 2n + 3$ for $\ell = 0$. There is one small isolated discrepancy in Table X, at $x_+ = 0.25$. [7] quotes $\Omega_{\rm K} = 2.41945 - 0.54735i$, whereas our spectral method gives $\Omega_{\rm SM} = 2.41841 - 0.54799i$. The difference, $|\Omega_{\rm SM} - \Omega_{\rm K}| \simeq 1.2 \times 10^{-3}$, is larger than a mere rounding effect. We repeated the computation and found the same value, stable under the changes of spectral resolution used in our convergence test. Since the neighbouring entries $x_+ = 0.3$ and $x_+ = 0.2$ agree with the published data to the displayed accuracy, and since the $x_+ = 0.25$ value lies smoothly on the small black hole trend toward the pure-AdS normal mode, we interpret this as an isolated discrepancy with the tabulated value in [7], rather than as a numerical instability of the present method.

Figure 1 provides a visual summary of the qualitative change between the small and large black hole regimes. For $x_+ = 0.01$, the imaginary parts are very small and the real parts are close to the pure-AdS sequence $3, 5, 7, \ldots$, confirming the weakly damped normal mode limit as $x_+ \to 0$. By contrast, for $x_+ = 50$ the modes are deeply damped and arrange themselves along two almost straight branches in the complex plane. The two branches are related by the symmetry $\Omega \mapsto -\Omega^*$, while the nearly constant spacing along each branch is the large black hole overtone behavior discussed above. Finally, Table XI shows how the scalar spectrum changes with the angular number ℓ in three representative regimes of the SAdS geometry. We display the first five modes, $n \in \{0, \ldots, 4\}$, for $x_+ \in \{0.4, 1, 50\}$. The $x_+ = 0.4$ panel illustrates the transition toward the small black hole regime. As ℓ increases, the real part of the fundamental mode increases and moves toward the pure AdS value $\Omega_{\rm AdS} = \ell + 3$, obtained from $\Omega_{\rm AdS} = 2n + \ell + 3$

at $n = 0$. The values $\Omega_0(\ell = 0) = 2.36293 - 1.00647i$, and $\Omega_0(\ell = 3) = 5.18494 - 0.27708i$, are consistent with this trend. At the same time, the damping rate decreases as ℓ increases. This is expected because the angular momentum barrier suppresses absorption by the black hole, making the modes longer lived. This behavior is the finite-x_+ version of the small black hole limit discussed in [7], where the imaginary part tends to zero and the real part approaches the normal-mode spectrum of pure AdS. For the intermediate case $x_+ = 1$, the same qualitative angular dependence is present, although the black hole scale and the AdS scale are comparable. The fundamental mode shifts from $\Omega_0 = 2.79822 - 2.67121i$ at $\ell = 0$ to $\Omega_0 = 5.00425 - 2.08948i$ at $\ell = 3$. Increasing ℓ therefore raises the oscillation frequency and reduces the damping of the fundamental mode. For the overtones, however, the dependence on ℓ becomes progressively weaker, since the spacing between successive modes approaches an approximately uniform pattern, consistent with the overtone behavior already observed in the $\ell = 0$ comparison tables. The large black hole panel, $x_+ = 50$, shows a qualitatively different feature. In this regime the spectrum is dominated by the horizon scale, and the dependence on ℓ is very weak. For example, the fundamental frequency changes only from $92.49368 - 133.19329i$ at $\ell = 0$ to $92.59332 - 133.16520i$ at $\ell = 3$, which is a small relative change compared with the overall scale of the frequency. The same weak angular dependence is visible for the first few overtones. This is consistent with the large SAdS-black hole scaling $\Omega_n \propto x_+$. The large black hole data also display the expected nearly uniform overtone spacing. For instance, for $x_+ = 50$ and $\ell = 0$, $\Omega_1 - \Omega_0 \simeq 65.60708 - 112.63110i$, while $\Omega_4 - \Omega_3 \simeq 65.01943 - 112.51774i$. Dividing by $x_+ = 50$, these spacings are already close to the asymptotic large black hole result $(\Omega_{n+1} - \Omega_n)/x_+ \simeq 1.299 - 2.25i$ reported in [8]. Thus, the angular number mainly affects the low-lying offset of the spectrum, whereas the high-overtone spacing, especially for large black holes, is controlled primarily by x_+ and only weakly depends on ℓ.

B. The scalar sector with generalized boundary condition at the conformal AdS boundary

We now turn to the generalized coefficient boundary condition $A\cos\theta + B\sin\theta = 0$. The endpoint $\theta = 0$ corresponds to the standard Dirichlet problem, while $\theta = \pi/2$ corresponds to the coefficient-Neumann condition. As emphasized above, this one-parameter family should be understood as a deformation of the minimally coupled massless scalar spectral problem, rather than as the standard alternative quantization available for scalars in the Breitenlohner–Freedman window. The main qualitative question is whether non-Dirichlet deformations preserve the stability of the scalar spectrum or generate additional modes with $\mathrm{Im}\,\Omega > 0$. Table XII summarizes the unstable mode found for three representative black hole sizes, $x_+ \in \{0.01, 1, 50\}$, and for several values of the boundary parameter θ. The Dirichlet endpoint $\theta = 0$ reproduces the standard scalar problem discussed in the previous subsection, and no unstable mode is found. By contrast, for every nonzero value of θ in the scan, an additional mode with $\mathrm{Im}\,\Omega > 0$ appears. With the time dependence $e^{-i\omega t}$, such a mode grows exponentially in time. The unstable frequency is accompanied by its symmetry related partner under $\Omega \mapsto -\Omega^*$. In the table we display only the branch with $\mathrm{Re}\,\Omega \geq 0$. The instability is present in all three regimes probed here. The case $x_+ = 0.01$ is close to the global-AdS limit, $x_+ = 1$ represents an intermediate size black hole, and $x_+ = 50$ lies deep in the large black hole regime. The appearance of the unstable branch in all three cases indicates that the effect is not tied to a particular black hole size, but rather to the non-Dirichlet deformation of the massless scalar boundary condition. The growth rate is very small for $x_+ = 0.01$, of order 10^{-5}, while it becomes of order unity for $x_+ = 1$ and of order x_+ for $x_+ = 50$. The values of $\Delta\Omega_{\max}$ show that the listed unstable modes are stable under the three-resolution matching procedure.

Figure 2 visualizes the dependence of the unstable branch on the boundary parameter. The growth rate remains positive for all nonzero values of θ shown, while the Dirichlet endpoint is stable. For $x_+ = 0.01$, the growth rate is very small and nearly independent of θ for $\theta \in \{\pi/8, \pi/4, 3\pi/8\}$, before increasing at the coefficient-Neumann endpoint. For $x_+ = 1$ and $x_+ = 50$, the same qualitative behavior is observed, i.e. the real part decreases as θ approaches $\pi/2$, and vanishes at the coefficient-Neumann endpoint, where the unstable mode becomes purely imaginary. The large black hole case has much larger values of both $\mathrm{Re}\,\Omega_{\mathrm{inst}}$ and $\mathrm{Im}\,\Omega_{\mathrm{inst}}$, reflecting the dominant scale set by the horizon radius. To determine whether the instability appears only after a finite deformation away from the Dirichlet endpoint, we performed a near-Dirichlet refinement. In particular, we considered $\theta \in \{\pi/64, \pi/32, \pi/16, (3\pi)/32, \pi/8\}$, again for $x_+ \in \{0.01, 1, 50\}$ and $\ell = 0$. The resulting unstable branch is shown in Fig. 3.

Figure 3 shows that the unstable mode persists throughout the near-Dirichlet refinement. We find an unstable mode for every nonzero value of θ tested, down to $\theta = \pi/64$, and for all three representative black hole sizes. Thus, no finite critical angle is observed in the interval $\pi/64 \leq \theta \leq \pi/8$. If a nonzero critical angle exists, it must satisfy $\theta_c < \pi/64$. The data are therefore consistent with the interpretation that the instability appears immediately once the Dirichlet condition is deformed. The behavior close to $\theta = 0$ depends on the black hole size. For the small black hole, $x_+ = 0.01$, the growth rate remains approximately constant, $\mathrm{Im}\,\Omega_{\mathrm{inst}} \simeq 3.19 \times 10^{-5}$, throughout the near-Dirichlet range shown. For $x_+ = 1$ and $x_+ = 50$, both $\mathrm{Re}\,\Omega_{\mathrm{inst}}$ and $\mathrm{Im}\,\Omega_{\mathrm{inst}}$ increase as θ is decreased. This indicates that the unstable branch does not continuously reduce to a finite-frequency Dirichlet QNM at $\theta = 0$. Rather, the

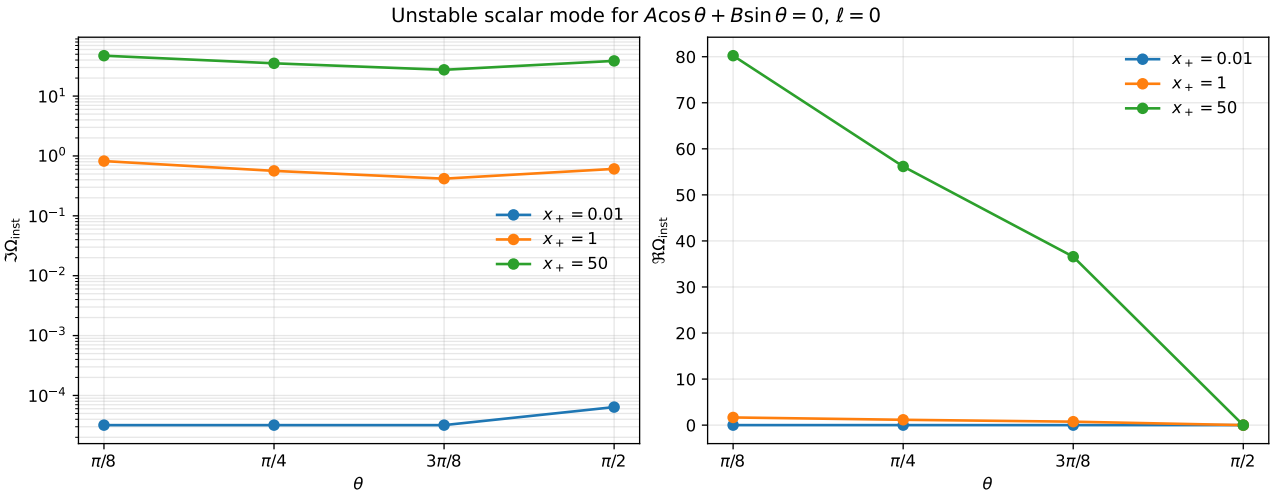


FIG. 2: *Unstable scalar mode for the generalized coefficient boundary condition* $A\cos\theta + B\sin\theta = 0$, *with* $\ell = 0$. *The left panel shows the growth rate* $\mathrm{Im}\,\Omega_{\rm inst}$, *while the right panel shows the oscillation frequency* $\mathrm{Re}\,\Omega_{\rm inst}$, *as functions of* θ. *The Dirichlet endpoint* $\theta = 0$ *is not shown because no unstable mode is found there. The growth-rate panel is shown on a logarithmic scale in order to display simultaneously the small black hole, intermediate, and large-black hole regimes.*

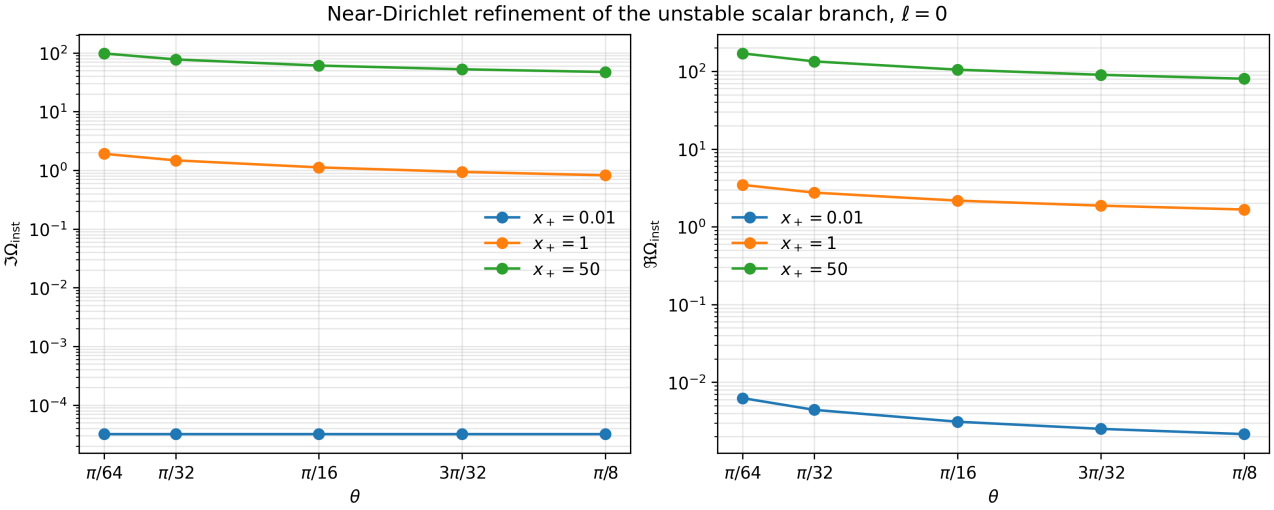


FIG. 3: *Near-Dirichlet refinement of the unstable scalar branch for the generalized coefficient boundary condition* $A\cos\theta + B\sin\theta = 0$, *with* $\ell = 0$. *The left panel shows the growth rate* $\mathrm{Im}\,\Omega_{\rm inst}$, *while the right panel shows the oscillation frequency* $\mathrm{Re}\,\Omega_{\rm inst}$. *Both vertical axes are shown on a logarithmic scale to display simultaneously the small, intermediate-, and large-black hole regimes. The Dirichlet endpoint* $\theta = 0$ *is not shown because no unstable mode is found there.*

standard Dirichlet problem appears as a special stable endpoint, while arbitrarily small non-Dirichlet deformations of the coefficient boundary condition introduce a separate unstable branch within the range probed by our numerical scan. These results show that the standard Dirichlet boundary condition is not merely a technical choice for the minimally coupled massless scalar. In the generalized coefficient problem considered here, deforming the relation between the slow and fast asymptotic coefficients changes the stability properties of the spectrum. The damped QNM tower remains present, but the new and physically most relevant feature is the appearance of an unstable mode for every nonzero value of θ considered.

TABLE VI: *Fundamental scalar modes across different SAdS black hole sizes x_+. We compare our spectral method results with $N = 200$ Chebyshev polynomials (final column) against the Horowitz–Hubeny power series method of [6] (second column), and the matrix method of [10] (third column) based on interpolation/discretization of the master equation. SM stands for spectral method.*

x_+	$\Omega_{\rm HH}$ [6]	$\Omega_{\rm LIN}$ [10]	$\Omega_{\rm SM}$
0.4	$2.3629 - 1.0064i$	$2.38152 - 0.938149i$	$2.3629 - 1.0065i$
0.6	$2.4316 - 1.5797i$	$2.4316 - 1.5797i$	$2.4316 - 1.5797i$
0.8	$2.5878 - 2.1304i$	$2.5878 - 2.1304i$	$2.5878 - 2.1304i$
1	$2.7982 - 2.6712i$	$2.7982 - 2.6712i$	$2.7982 - 2.6712i$
5	$9.4711 - 13.3255i$	$9.4711 - 13.3255i$	$9.4711 - 13.3255i$
10	$18.6070 - 26.6418i$	$18.6070 - 26.6418i$	$18.6070 - 26.6418i$
50	$92.4937 - 133.1933i$	$92.4937 - 133.1933i$	$92.4937 - 133.1933i$
100	$184.9534 - 266.3856i$	$184.9534 - 266.3856i$	$184.9534 - 266.3856i$

V. CONCLUSIONS AND OUTLOOK

In this work, we have studied QNMs of a minimally coupled massless scalar field on four-dimensional SAdS–black holes. The analysis was organized around two related spectral problems. The first is the standard scalar SAdS problem, defined by purely ingoing behavior at the event horizon and the Dirichlet, or vanishing field, condition at the conformal AdS boundary. The second is a generalized coefficient problem in which the two independent asymptotic data of the same massless scalar are related by a one-parameter boundary condition interpolating between the Dirichlet endpoint and a coefficient-Neumann endpoint. On the numerical side, we developed a Chebyshev spectral implementation in which the radial equation is compactified to a finite interval and recast as a quadratic matrix pencil in the dimensionless frequency $\Omega = \omega R$. For the Dirichlet problem, both the ingoing horizon behavior and the AdS-boundary falloff are incorporated directly into the radial ansatz. For the generalized coefficient problem, the ingoing horizon behavior is factored out, while the boundary condition at the conformal AdS boundary is imposed as a spectral boundary row. In the latter case, the boundary row makes the leading quadratic matrix singular, and we therefore implemented the problem by eliminating the boundary constraint before solving the reduced quadratic pencil. In both cases, physical modes were extracted by matching spectra obtained at the three resolutions $N \in \{190, 195, 200\}$, and only modes stable under this triplet procedure were retained. For the standard Dirichlet scalar problem, the method reproduces the known Schwarzschild–AdS scalar spectrum with high accuracy. We compared against the Horowitz–Hubeny power-series method, Konoplya's small-black-hole data, the Cardoso–Konoplya–Lemos overtone tables, the asymptotic iteration method, the Lin–Qian matrix method, and the continued fraction results of Daghigh, Green and Morey. The agreement across these independent methods provides a stringent validation of the spectral construction. The calculations also reproduce the expected limiting behaviors. In the small black hole regime, the scalar frequencies approach the pure AdS normal mode sequence and the damping rate tends to zero. In the large black hole regime, the frequencies scale linearly with the horizon radius and the highly damped modes approach the known nearly uniform spacing in the complex-frequency plane. The angular-momentum scan further confirms that ℓ strongly affects the low lying modes for small and intermediate black holes, while its relative effect becomes weak in the large black hole regime, where the dominant scale is set by $x_+ = r_+/R$. The generalized coefficient boundary condition leads to a qualitatively different result. The Dirichlet endpoint reproduces the stable standard spectrum. However, for every nonzero value of the boundary parameter examined, we find an additional mode with positive imaginary part. With the convention $e^{-i\omega t}$, this corresponds to exponential growth in time. The unstable mode appears together with its symmetry-related partner under $\Omega \mapsto -\Omega^*$, and it is present for small, intermediate, and large black holes. A near-Dirichlet refinement shows that the unstable branch persists down to the smallest deformation we probed, $\theta = \pi/64$. Thus, no finite critical angle was observed in our numerical scan. We do not claim this as a proof that the instability exists for every arbitrarily small nonzero deformation, but the data are consistent with the interpretation that the Dirichlet endpoint is a special stable point of the generalized coefficient problem. It is important to stress the interpretation of this result. The generalized boundary condition studied here is not the usual alternative quantization of a scalar field in the Breitenlohner–Freedman window. It is instead a controlled deformation of the minimally coupled massless scalar QNM problem, formulated directly in terms of the independent asymptotic coefficients of the massless field. From this viewpoint, the instability shows that the standard Dirichlet prescription

TABLE VII: *Detailed overtone comparison at $x_+ = 1$ and $\ell = 0$. We compare our spectral method results, obtained with $N = 200$ Chebyshev polynomials and checked against $N = 190, 195$, with the data obtained by means of the Horowitz–Hubeny power series method [8] and the asymptotic iteration method of [9] with 70 iteration. Entries labelled N/A indicate unavailable data. SM stands for spectral method. †: The value $8.46153 - 9.74852i$ appearing in [8] for this mode is a known mismatch in the real part and it was already reported by [9].*

n	$\Omega_{\rm CKL}$ [8]	$\Omega_{\rm AIM}$ [9]	$\Omega_{\rm SM}$
0	$2.7982 - 2.6712i$	$2.79823 - 2.67121i$	$2.7982 - 2.6712i$
1	$4.75849 - 5.03757i$	$4.75850 - 5.03757i$	$4.7585 - 5.0376i$
2	$6.71927 - 7.39449i$	$6.71931 - 7.39450i$	$6.7193 - 7.3945i$
3	$8.46153^{\dagger} - 9.74852i$	$8.68233 - 9.74854i$	$8.6822 - 9.7485i$
4	$10.6467 - 12.1012i$	$10.6469 - 12.1013i$	$10.6467 - 12.1012i$
5	$12.6121 - 14.4533i$	$12.6125 - 14.4533i$	$12.6121 - 14.4533i$
6	$14.5782 - 16.8049i$	$14.5788 - 16.8050i$	$14.5782 - 16.8049i$
7	$16.5449 - 19.1562i$	$16.5457 - 19.1563i$	$16.5449 - 19.1562i$
8	$18.5119 - 21.5073i$	N/A	$18.5119 - 21.5073i$
9	$20.4792 - 23.8583i$	N/A	$20.4792 - 23.8583i$
10	$22.44671 - 26.20913i$	N/A	$22.4467 - 26.2091i$
11	$24.41443 - 28.55989i$	N/A	$24.4144 - 28.5599i$
12	$26.38230 - 30.91059i$	N/A	$26.3823 - 30.9106i$
13	$28.35029 - 33.26123i$	N/A	$28.3503 - 33.2612i$
14	$30.31839 - 35.61183i$	N/A	$30.3184 - 35.6118i$
15	$32.28658 - 37.96238i$	N/A	$32.2866 - 37.9624i$
16	$34.25485 - 40.31290i$	N/A	$34.2548 - 40.3129i$
17	$36.22318 - 42.66340i$	N/A	$36.2232 - 42.6634i$
18	$38.19157 - 45.01387i$	N/A	$38.1916 - 45.0139i$
19	$40.16002 - 47.36431i$	N/A	$40.1600 - 47.3643i$
20	N/A	N/A	$42.1285 - 49.7147i$

is not merely a harmless technical choice because changing the relation between the slow and fast asymptotic data can alter the stability properties of the spectrum. This observation is consistent with the broader lesson from AdS perturbation theory, namely that the boundary condition at the conformal boundary is an intrinsic part of the physical spectral problem.

Several extensions are natural. The first is to understand analytically the origin of the unstable branch. In particular, it would be useful to determine whether the branch can be followed perturbatively away from the Dirichlet endpoint, whether its frequency diverges or leaves the physical sheet as $\theta \to 0$, and whether one can establish a rigorous instability criterion for the generalized coefficient boundary condition. A complementary time-domain study would also be valuable. The frequency domain calculation identifies exponentially growing modes, but direct evolution would show how these modes dominate generic initial data and would clarify the role of the boundary condition in the associated energy balance. A second direction is to repeat the analysis for scalar fields whose two AdS falloffs are both admissible in the standard sense. In particular, conformally coupled or massive scalars in the appropriate mass range admit genuine Robin or mixed boundary conditions related to the usual alternative quantization. Studying those cases with the same spectral machinery would allow a clean comparison between the generalized coefficient deformation considered here and physically admissible Robin families in the Ishibashi–Wald framework. It would also make contact with recent work on Robin boundary conditions and boundary condition induced instabilities in AdS black-hole backgrounds. A third and immediate extension is to apply the present spectral method to the electromagnetic and gravitational sectors. This is particularly important because boundary conditions for higher-spin perturbations in AdS are known to be more subtle than simple field vanishing. For Maxwell perturbations, vanishing

TABLE VIII: *Detailed overtone comparison at $x_+ \in \{0.2, 100\}$ and $\ell = 0$. We compare our spectral method results, obtained with $N = 200$ Chebyshev polynomials and checked against $N = 190, 195$, with the data obtained by means of the Horowitz–Hubeny power series method [8]. Entries labelled N/A indicate unavailable data. SM stands for spectral method.*

x_+	n	Ω_{CKL} [8]	Ω_{SM}	x_+	n	Ω_{CKL} [8]	Ω_{SM}
0.2	0	$2.47511 - 0.38990i$	$2.4751 - 0.3899i$	100	0	$184.95344 - 266.38560i$	$184.9534 - 266.3856i$
	1	$4.07086 - 0.98966i$	$4.0709 - 0.9866i$		1	$316.14466 - 491.64354i$	$316.1447 - 491.6435i$
	2	$5.72783 - 1.57600i$	$5.7278 - 1.5760i$		2	$446.46153 - 716.75722i$	$446.4615 - 716.7572i$
	3	$7.40091 - 2.15869i$	$7.4009 - 2.1587i$		3	$576.55983 - 941.81253i$	$576.5598 - 941.8125i$
	4	$9.08118 - 2.73809i$	$9.0812 - 2.7381i$		4	$706.57518 - 1166.8440i$	$706.5752 - 1166.8441i$
	5	$10.7655 - 3.31557i$	$10.7655 - 3.3156i$		5	$836.55136 - 1391.8641i$	$836.5514 - 1391.8641i$
	6	$12.45222 - 3.89179i$	$12.4522 - 3.8918i$		6	$966.50635 - 1616.8779i$	$966.5063 - 1616.8780i$
	7	$14.14065 - 4.46714i$	$14.1406 - 4.4671i$		7	$1096.44876 - 1841.88813i$	$1096.4488 - 1841.8881i$
	8	$15.83026 - 5.04186i$	$15.8303 - 5.0419i$		8	$1226.38317 - 2066.89596i$	$1226.3832 - 2066.8960i$
	9	$17.52070 - 5.61610i$	$17.5207 - 5.6161i$		9	$1356.31222 - 2291.90222i$	$1356.3122 - 2291.9022i$
	10	$19.21191 - 6.18997i$	$19.2119 - 6.1900i$		10	$1486.23753 - 2516.90740i$	$1486.2375 - 2516.9074i$
	11	$20.90359 - 6.76355i$	$20.9036 - 6.7635i$		11	N/A	$1616.1601 - 2741.9118i$
	12	N/A	$22.5957 - 7.3369i$		12	N/A	$1746.0808 - 2966.9156i$
	13	N/A	$24.2881 - 7.9100i$		13	N/A	$1875.9998 - 3191.9190i$
	14	N/A	$25.9809 - 8.4830i$		14	N/A	$2005.9177 - 3416.9220i$
	15	N/A	$27.6738 - 9.0558i$		15	N/A	$2135.8347 - 3641.9248i$
	16	N/A	$29.3670 - 9.6285i$		16	N/A	$2265.7509 - 3866.9274i$
	17	N/A	$31.0603 - 10.2011i$		17	N/A	$2395.6665 - 4091.9298i$
	18	N/A	$32.7538 - 10.7737i$		18	N/A	$2525.5817 - 4316.9320i$
	19	N/A	$34.4474 - 11.3461i$		19	N/A	$2655.4964 - 4541.9342i$
	20	N/A	$36.1412 - 11.9185i$		20	N/A	$2785.4107 - 4766.9362i$

TABLE IX: *Independent continued-fraction benchmark for the scalar sector at $x_+ = 50$ and $\ell = 0$. We compare our spectral method frequencies obtained with $N = 200$ Chebyshev polynomials and checked against $N = 190, 195$ with the continued fraction results of [11]. Their overtone label starts at $n = 1$. Here we relabel the fundamental mode as $n = 0$. SM stands for spectral method.*

n	Ω_{DGM} [11]	Ω_{SM}
0	$92.4936786 - 133.1932905i$	$92.4937 - 133.1933i$
1	$158.1007632 - 245.8243897i$	$158.1008 - 245.8244i$
2	$223.2708309 - 358.3832592i$	$223.2708 - 358.3833i$
3	$288.3316967 - 470.9129117i$	$288.3317 - 470.9129i$
4	$353.3511287 - 583.4306502i$	$353.3511 - 583.4307i$
5	$418.3509995 - 695.9426546i$	$418.3510 - 695.9427i$

energy flux at the AdS boundary permits more than one reflecting branch, and the second branch gives spectra distinct from the older Dirichlet-type calculations. For gravitational perturbations, the axial and polar sectors require special care because fixed boundary metric or AdS/CFT-motivated prescriptions lead naturally to parity-dependent boundary conditions, with Dirichlet-type behavior in the axial sector and Robin-type behavior in the polar sector. The matrix-pencil formulation developed here is well suited to these problems, since Dirichlet, Neumann, Robin, and

TABLE X: *Small black hole scalar spectrum for $\ell = 0$ and several values of x_+. We compare the fundamental mode obtained with our spectral method with the small black hole data of [7]. The latter were obtained using a Horowitz–Hubeny-type power-series expansion adapted to the small x_+ regime. Our results were computed with $N = 200$ Chebyshev polynomials and checked against $N = 190, 195$. Entries labelled N/A indicate unavailable data, while SM stands for spectral method.*

x_+	$\Omega_{\rm K}$ $(n=0)$ [7]	$\Omega_{\rm SM}$ $(n=0)$	$\Omega_{\rm SM}$ $(n=1)$	$\Omega_{\rm SM}$ $(n=2)$	$\Omega_{\rm SM}$ $(n=3)$
0.3	$2.38447 - 0.70413i$	$2.3845 - 0.7041i$	$3.9836 - 1.5467i$	$5.6193 - 2.3692i$	$7.2658 - 3.1857i$
0.25	$2.41945 - 0.54735i$	$2.4184 - 0.5480i$	$4.0141 - 1.2726i$	$5.6571 - 1.9800i$	$7.3131 - 2.6811i$
0.2	$2.47511 - 0.3899i$	$2.4751 - 0.3899i$	$4.0709 - 0.9866i$	$5.7278 - 1.5760i$	$7.4009 - 2.1587i$
0.125	$2.62274 - 0.16392i$	$2.6227 - 0.1639i$	$4.2386 - 0.5247i$	$5.9267 - 0.9242i$	$7.6426 - 1.3201i$
0.1	$2.69282 - 0.10096i$	$2.6928 - 0.1010i$	$4.3352 - 0.3626i$	$6.0351 - 0.6876i$	$7.7695 - 1.0168i$
1/12	$2.74472 - 0.06616i$	$2.7447 - 0.0662i$	$4.4189 - 0.2567i$	$6.1288 - 0.5232i$	$7.8759 - 0.8045i$
1/14	$2.78341 - 0.04578i$	$2.7835 - 0.0458i$	$4.4902 - 0.1857i$	$6.2114 - 0.4037i$	$7.9674 - 0.6470i$
1/16	$2.81289 - 0.03311i$	$2.8129 - 0.0331i$	$4.5501 - 0.1373i$	$6.2847 - 0.3147i$	$8.0480 - 0.5258i$
1/18	$2.83574 - 0.02491i$	$2.8357 - 0.0249i$	$4.6001 - 0.1038i$	$6.3498 - 0.2477i$	$8.1202 - 0.4305i$
1/20	$2.8539 - 0.01932i$	$2.8539 - 0.0193i$	$4.6418 - 0.0802i$	$6.4075 - 0.1969i$	$8.1854 - 0.3544i$
1/25	$2.88584 - 0.01138i$	$2.8859 - 0.0114i$	$4.7191 - 0.0458i$	$6.5233 - 0.1161i$	$8.3233 - 0.2231i$
1/30	$2.9065 - 0.0074i$	$2.9065 - 0.0075i$	$4.7707 - 0.0289i$	$6.6071 - 0.0732i$	$8.4311 - 0.1455i$
1/100	N/A	$2.9738 - 0.0006i$	$4.9378 - 0.0018i$	$6.8953 - 0.0038i$	$8.8475 - 0.0069i$

TABLE XI: *First five scalar modes for selected values of the black hole size x_+ and angular number ℓ. We use the standard vanishing field boundary condition at the AdS boundary. The frequencies were obtained with the spectral method using $N = 200$ Chebyshev polynomials and checked against $N = 190, 195$.*

$x_+ = 0.4$					
ℓ	Ω_0	Ω_1	Ω_2	Ω_3	Ω_4
0	$2.3629 - 1.0065i$	$3.9786 - 2.0728i$	$5.6171 - 3.1196i$	$7.2637 - 4.1611i$	$8.9145 - 5.2002i$
1	$3.1338 - 0.7029i$	$4.5679 - 1.7813i$	$6.1146 - 2.8525i$	$7.7035 - 3.9152i$	$9.3136 - 4.9717i$
2	$4.1285 - 0.4512i$	$5.3977 - 1.4466i$	$6.8358 - 2.5038i$	$8.3492 - 3.5712i$	$9.9042 - 4.6378i$
3	$5.1849 - 0.2771i$	$6.3383 - 1.1521i$	$7.6794 - 2.1645i$	$9.1155 - 3.2154i$	$10.6096 - 4.2786i$
$x_+ = 1$					
ℓ	Ω_0	Ω_1	Ω_2	Ω_3	Ω_4
0	$2.7982 - 2.6712i$	$4.7585 - 5.0376i$	$6.7193 - 7.3945i$	$8.6822 - 9.7485i$	$10.6467 - 12.1013i$
1	$3.3306 - 2.4895i$	$5.1724 - 4.8881i$	$7.0710 - 7.2646i$	$8.9940 - 9.6321i$	$10.9298 - 11.9948i$
2	$4.1171 - 2.2756i$	$5.8165 - 4.6678i$	$7.6311 - 7.0561i$	$9.4978 - 9.4367i$	$11.3927 - 11.8112i$
3	$5.0043 - 2.0895i$	$6.5791 - 4.4385i$	$8.3067 - 6.8206i$	$10.1116 - 9.2058i$	$11.9603 - 11.5878i$
$x_+ = 50$					
ℓ	Ω_0	Ω_1	Ω_2	Ω_3	Ω_4
0	$92.4937 - 133.1933i$	$158.1008 - 245.8244i$	$223.2708 - 358.3833i$	$288.3317 - 470.9129i$	$353.3511 - 583.4307i$
1	$92.5103 - 133.1886i$	$158.1134 - 245.8211i$	$223.28141 - 358.3805i$	$288.3410 - 470.9105i$	$353.3595 - 583.4285i$
2	$92.5435 - 133.1792i$	$158.1387 - 245.8144i$	$223.3026 - 358.3749i$	$288.3595 - 470.9056i$	$353.3762 - 583.4241i$
3	$92.5933 - 133.1652i$	$158.1765 - 245.8044i$	$223.3343 - 358.3666i$	$288.3873 - 470.8983i$	$353.4013 - 583.4175i$

more general coefficient conditions can all be incorporated at the level of spectral boundary rows. Beyond spherical Schwarzschild–AdS, it would also be interesting to consider charged, rotating, or topological AdS black holes. In those settings, one expects a richer interplay between horizon physics, boundary conditions, and possible instabilities. Another promising direction is to use the high precision spectra obtained here to investigate non-normality and pseudospectral sensitivity of AdS QNM problems. Since the QNM operator is non-self-adjoint, small changes in the boundary condition or in the operator coefficients can reorganize the spectrum in ways not captured by eigenvalues alone. The generalized boundary condition instability found in this work provides a concrete setting in which such questions can be explored.

TABLE XII: *Unstable scalar mode for the generalized coefficient boundary condition* $A\cos\theta + B\sin\theta = 0$*, with* $\ell = 0$*. We display only the branch with* $\mathrm{Re}\,\Omega \geq 0$*. The partner with negative real part is obtained from* $\Omega \mapsto -\Omega^*$*. The entry "none" at* $\theta = 0$ *corresponds to the standard Dirichlet problem, for which no unstable mode is found. The frequencies were obtained with the spectral method using* $N = 200$ *Chebyshev polynomials and checked against* $N = 190, 195$*. Here* $\Delta\Omega_{\max}$ *denotes the largest pairwise difference between the frequencies obtained at the three spectral resolutions* $N \in \{190, 195, 200\}$*.*

x_+	θ	$\mathrm{Re}\,\Omega_{\mathrm{inst}}$	$\mathrm{Im}\,\Omega_{\mathrm{inst}}$	$\Delta\Omega_{\max}$
0.01	0	none	none	–
0.01	$\pi/8$	0.0022	0.0000	2.76×10^{-11}
0.01	$\pi/4$	0.0014	0.0000	1.78×10^{-11}
0.01	$3\pi/8$	0.0009	0.0000	1.15×10^{-11}
0.01	$\pi/2$	0.0000	0.0001	3.12×10^{-12}
1	0	none	none	–
1	$\pi/8$	1.6658	0.8213	1.18×10^{-106}
1	$\pi/4$	1.1590	0.5634	2.82×10^{-107}
1	$3\pi/8$	0.7483	0.4174	1.01×10^{-108}
1	$\pi/2$	0.0000	0.6081	2.15×10^{-107}
50	0	none	none	–
50	$\pi/8$	80.2260	47.2803	2.89×10^{-127}
50	$\pi/4$	56.1688	35.2695	7.81×10^{-128}
50	$3\pi/8$	36.5749	27.4649	2.02×10^{-128}
50	$\pi/2$	0.0000	38.6960	2.21×10^{-128}

Code availability

The codes used to assemble the spectral matrices and to post-process the quadratic eigenvalue problems are available at the following URL address:

https://github.com/dutykh/schwarzschild-ads/

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